

Exercise (1) – The complete list of problems can be found in the book.

2. (a) What is the maximum allowable current through a 10 kΩ, 10 W resistor?
 (b) Through a 10 kΩ, 1/4 W resistor?

Solution:

$$R := 10 \text{ k}\Omega$$

$$P_a := 10 \text{ W}$$

$$P_b := \frac{1}{4} \text{ W}$$

$$\text{As } P := I^2 \cdot R \rightarrow$$

$$I_a := \sqrt{\frac{P_a}{R}} = 31.623 \text{ mA}$$

$$I_b := \sqrt{\frac{P_b}{R}} = 5 \text{ mA}$$

3. (a) What power rating is needed for a 100 Ω resistor if 100 V is to be applied to it?
 (b) For a 100 kΩ resistor?

Solution:

$$U := 100 \text{ V}$$

$$R_a := 100 \Omega$$

$$R_b := 100 \text{ k}\Omega$$

$$\text{As } P := \frac{U^2}{R} \rightarrow$$

$$P_a := \frac{U^2}{R_a} = 100 \text{ W}$$

$$P_b := \frac{U^2}{R_b} = 0.1 \text{ W}$$

6. The output of the voltage divider is to be measured with voltmeters with input resistances of 100 Ω, 1 kΩ, 50 kΩ, and 1 MΩ. What voltage will each indicate?

Solution:

Write **Kirchoff's Current Law** (law of **nodes**) for the V_{out} node with **voltages** across each component, including meter-resistance:

$$0 = I_2 - I_1 - I_m = \frac{U_2}{R_2} - \frac{U_1}{R_1} - \frac{U_m}{R_m}$$

Identify voltages as **potential-differences** (referenced to the 0 V wire):

$$U_2 = V_{\text{in}} - V_{\text{out}}, \quad U_1 = U_m = V_{\text{out}} - 0 \text{ V}$$

Rearrange currents:

$$0 = \frac{V_{\text{in}} - V_{\text{out}}}{R_2} - \frac{V_{\text{out}}}{R_1} - \frac{V_{\text{out}}}{R_m} = \frac{V_{\text{in}}}{R_2} - V_{\text{out}} \left(\frac{1}{R_2} + \frac{1}{R_1} + \frac{1}{R_m} \right)$$

$$V_{\text{out}} = \frac{V_{\text{in}}}{R_2} \left(\frac{1}{R_2} + \frac{1}{R_1} + \frac{1}{R_m} \right)^{-1}$$

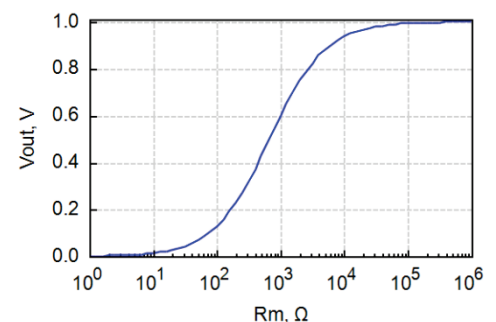
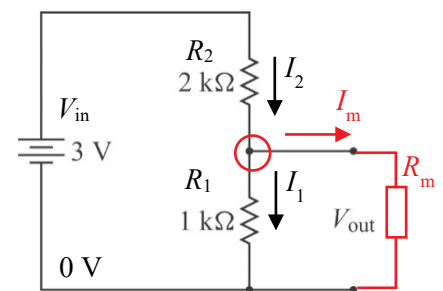
Insert values as a function of meter-resistance:

$$R_1 := 1 \text{ k}\Omega$$

$$R_2 := 2 \text{ k}\Omega$$

$$V_{\text{in}} := 3 \text{ V}$$

$$V_{\text{out}}(R_m) := \frac{V_{\text{in}}}{R_2} \cdot \left(\frac{1}{R_2} + \frac{1}{R_1} + \frac{1}{R_m} \right)^{-1}$$



$$V_{\text{out}}(100 \Omega) = 0.13 \text{ V} \quad V_{\text{out}}(1 \text{ k}\Omega) = 0.6 \text{ V} \quad V_{\text{out}}(50 \text{ k}\Omega) = 0.987 \text{ V} \quad V_{\text{out}}(1 \text{ M}\Omega) = 0.999 \text{ V}$$

7. A real battery can be modelled as an ideal voltage source in series with a resistor (the **internal resistance**). A voltmeter with input resistance of $1000\ \Omega$ measures the voltage of a worn-out 1.5 V flashlight battery as 0.9 V . What is the internal resistance of the battery?

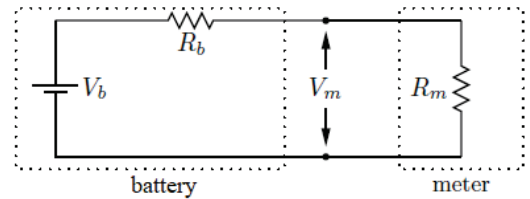
Solution:

When the voltmeter is attached, the circuit appears as a voltage divider, so:

$$V_m = V_b \frac{R_m}{R_b + R_m} \rightarrow V_m R_b + V_m R_m = V_b R_m$$

$$V_b := 1.5\text{ V} \quad V_m := 0.9\text{ V} \quad R_m := 1000\ \Omega$$

$$R_b := \frac{R_m}{V_m} \cdot (V_b - V_m) = 666.667\ \Omega$$



13. Find the Thevenin voltage and Thevenin resistance of the circuit shown with R_5 removed. The two terminals for this problem are the points where R_5 was connected.

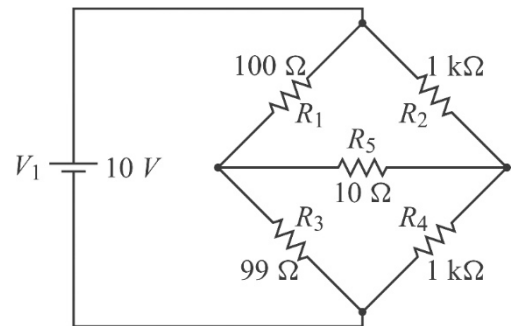
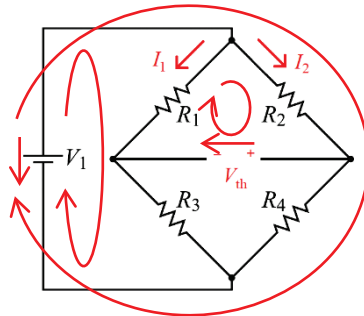
Solution:

Assume V_{th} polarity and loops as drawn. From outer to inner loops:

$$0 = I_2(R_2 + R_4) - V_1$$

$$0 = I_1(R_1 + R_3) - V_1$$

$$0 = I_2 R_2 - I_1 R_1 + V_{th}$$



Expressing currents:

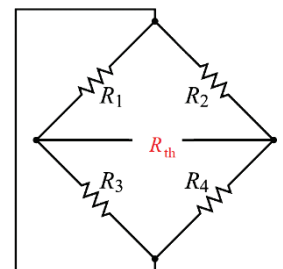
$$I_2 = \frac{V_1}{R_2 + R_4}, \quad I_1 = \frac{V_1}{R_1 + R_3}, \quad V_{th} = I_1 R_1 - I_2 R_2 = V_1 \left(\frac{R_1}{R_1 + R_3} - \frac{R_2}{R_2 + R_4} \right)$$

$$R_1 := 100\ \Omega \quad R_2 := 1\text{ k}\Omega \quad R_3 := 99\ \Omega \quad R_4 := R_2 \quad R_5 := 10\ \Omega \quad V_1 := 10\text{ V}$$

$$V_{th} := V_1 \cdot \left(\frac{R_1}{R_1 + R_3} - \frac{R_2}{R_2 + R_4} \right) = 0.025\text{ V}$$

To find R_{th} , voltage source V_1 is shorted. Now $R_1 \parallel R_3$ and $R_2 \parallel R_4$. The resultants will be in series:

$$R_{th} := \frac{R_1 \cdot R_3}{R_1 + R_3} + \frac{R_2 \cdot R_4}{R_2 + R_4} = 549.749\ \Omega$$



14. Using the result of the previous problem, find the current through R_5 of the pervious circuit (*Problem-1.13*):

Solution:

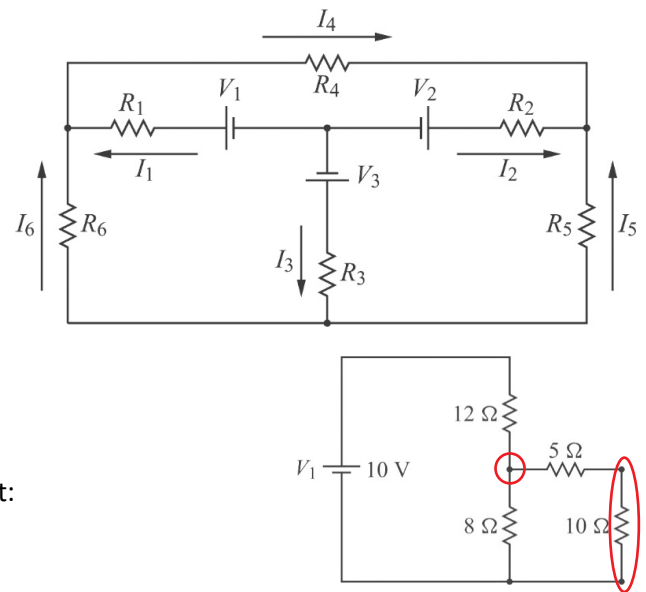
The Thevenin resistance is in series with V_{th} , hence it will be in series with R_5 :

$$I_5 := \frac{V_{th}}{R_{th} + R_5} = 4.489 \cdot 10^{-5}\text{ A} \quad I_5 = 44.887\ \mu\text{A}$$

18. (a) Compute the current through the $10\ \Omega$ resistor in the circuit. Do not use Thevenin's or Norton's theorems for this computation.

(b) Now find the Thevenin voltage, the Thevenin resistance, and the Norton current when the $10\ \Omega$ resistor is removed. The two terminals for this problem are the points where the $10\ \Omega$ resistor is connected.

(c) Show that, if the $10\ \Omega$ resistor is connected to the Thevenin equivalent circuit, the current through the $10\ \Omega$ resistor matches the value found in part (a). Do the same for the Norton equivalent circuit.



Solution:

(a) Calculate first the total resistance, then the total current:

$$R_{\text{tot}} = 12\ \Omega + 8\ \Omega \parallel (5\ \Omega + 10\ \Omega)$$

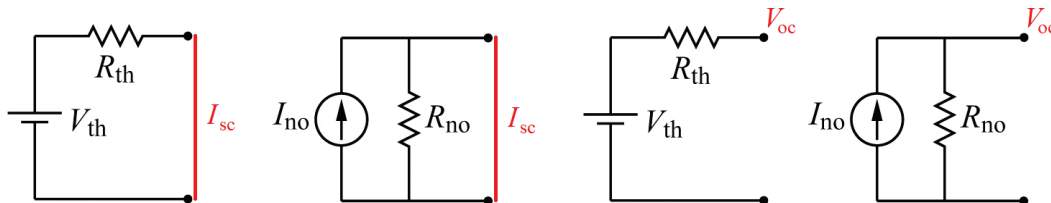
$$R_{\text{tot}} := 12\ \Omega + \left(\frac{1}{8\ \Omega} + \frac{1}{5\ \Omega + 10\ \Omega} \right)^{-1} = 17.217\ \Omega \quad I_{\text{tot}} := \frac{10\ \text{V}}{R_{\text{tot}}} = 0.581\ \text{A}$$

The current is calculated from the node voltage $V_{8\Omega}$ of the circled node:

$$V_{8\Omega} := V_1 - 12\ \Omega \cdot I_{\text{tot}} = 3.03\ \text{V} \quad I_{10\Omega} := \frac{V_{8\Omega}}{5\ \Omega + 10\ \Omega} = 0.202\ \text{A}$$

(b) The Thevenin voltage is the voltage across the terminals with **no current** through the $5\ \Omega$ resistor. The Thevenin resistance is the resultant if we short the voltage source.

$$V_{\text{th}} := 10\ \text{V} \cdot \frac{8\ \Omega}{8\ \Omega + 12\ \Omega} = 4\ \text{V} \quad R_{\text{th}} := 5\ \Omega + \left(\frac{1}{8\ \Omega} + \frac{1}{12\ \Omega} \right)^{-1} = 9.8\ \Omega$$



The Norton current will flow if we **short** the terminals of both equivalents (left two figures). From the Thevenin equivalent, the short circuit current is $I_{\text{no}} = I_{\text{sc}} = \frac{V_{\text{th}}}{R_{\text{th}}}$. Similarly, the Thevenin voltage will be present at the terminals of both equivalents, if we leave them **open**: $V_{\text{th}} = V_{\text{oc}} = R_{\text{no}} I_{\text{no}}$. From these, it is clear that the two resultant resistances are the same: $I_{\text{no}} = \frac{R_{\text{no}} I_{\text{no}}}{R_{\text{th}}}$, giving the Norton current as:

$$I_{\text{no}} := \frac{V_{\text{th}}}{R_{\text{th}}} = 0.408\ \text{A}$$

(c) In the Thevenin equivalent, the $10\ \Omega$ is in **series**:

$$I_{10\Omega} := \frac{V_{\text{th}}}{R_{\text{th}} + 10\ \Omega} = 0.202\ \text{A}$$

In the Norton equivalent, the $10\ \Omega$ is in parallel, yielding a **current divider**:

$$I_{R_{\text{no}}} R_{\text{no}} = I_{10\Omega} 10\ \Omega, \quad I_{\text{no}} = I_{R_{\text{no}}} + I_{10\Omega} = \frac{I_{10\Omega} 10\ \Omega}{R_{\text{no}}} + I_{10\Omega} = I_{10\Omega} \left(1 + \frac{10\ \Omega}{R_{\text{th}}} \right)$$

$$I_{10\Omega} := \frac{I_{\text{no}}}{1 + \frac{10\ \Omega}{R_{\text{th}}}} = 0.202\ \text{A}$$